

The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

The marginal effect of increasing the house size (SQFT) is:

$$\frac{\partial PRICE}{\partial SQFT} = 2\alpha_2 \cdot SQFT$$

At 2000 square feet ($SQFT = 20$), the marginal effect is: $ME = 2\alpha_2 \cdot 20 = 40\alpha_2$

The hypotheses are:

$$H_0: ME \leq 13 \Rightarrow \alpha_2 \leq 0.325$$

$$H_1: ME > 13 \Rightarrow \alpha_2 > 0.325$$

The test statistic is

$$t = \frac{\hat{\alpha}_2 - 0.325}{se(\hat{\alpha}_2)} \cong -26.72875$$

At the 5% significance level, the critical value is approximately 1.647919. The rejection region is: $t > 1.647919$. Since $-26.72875 < 1.647919$, we fail to reject the null hypothesis. The p-value is approximately 1, which is greater than 0.05, so we fail to reject H_0 .

Fail to reject the null hypothesis. There is insufficient evidence at the 5% significance level to conclude that the marginal effect of increasing a 2000 square foot house by 100 square feet exceeds \$13000.

b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

At 4000 square feet ($SQFT = 40$), the marginal effect is: $ME = 2\alpha_2 \cdot 40 = 80\alpha_2$

The hypotheses are:

$$H_0: ME \leq 13 \Rightarrow \alpha_2 \leq 0.1625$$

$$H_1: ME > 13 \Rightarrow \alpha_2 > 0.1625$$

The test statistic is

$$t = \frac{\hat{\alpha}_2 - 0.1625}{se(\hat{\alpha}_2)} \cong 4.189467$$

At the 5% significance level, the critical value is approximately 1.647919. The rejection region is: $t > 1.647919$. Since $4.189467 > 1.647919$, we reject the null hypothesis. The p-value is approximately 1.654×10^{-5} , which is smaller than 0.05, so we reject H_0 .

Reject the null hypothesis. There is sufficient evidence at the 5% significance level to conclude that the marginal effect of increasing a 4000 square foot house by 100 square feet exceeds \$13,000.

c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

$$E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$$

At 2000 square feet ($SQFT = 20$), the expected price is: $\alpha_1 + (20)^2 = \alpha_1 + 400\alpha_2$

Using the regression model, the estimated expected price is 167.3735. At 95% confidence interval for the expected price is [158.0481, 176.6988].

We are 95% confident that the true average price of houses with 2000 square feet lies within this interval. This interval reflects uncertainty in estimating the average price, not the price of an individual house.

d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

The sample average price for houses with 2000 square feet is 163.333 (in thousands of dollars).

Comparing this with the 95% confidence interval obtained in part (c), which is [158.0481, 176.6988], we find that the sample mean lies within this interval.

Therefore, the sample average is compatible with the regression result, suggesting that the model provides a reasonable estimate of the average price.